

2023 AMC 10A Problem 1 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 1. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 1

Problem:

Cities A and B are 45 miles apart. Alicia lives in A and Beth lives in B . Alicia bikes towards B at 18 miles per hour. Leaving at the same time, Beth bikes toward A at 12 miles per hour. How many miles from City A will they be when they meet?

Answer Choices:

- A. 20
- B. 24
- C. 25
- D. 26
- E. 27

Solution:

Let t be the number of hours they ride before meeting. Then $18t + 12t = 45$. This gives $t = 1.5$, so they each ride for 1.5 hours. Then Alicia has biked $18 \cdot 1.5 = 27$ miles, so their meeting point is (E) 27 miles from A .

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) [↗](#)

